

JunYoung Ahn

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Education

Seoul National University

Bachelor of Science in Mechanical Engineering

Seoul, Republic of Korea / [Mar 2024 – Expected Feb 2028]

Relevant Coursework: Advanced Topics in Mechanical Engineering 1, Field Application of Blockchain 1, Engineering Vibration, Dynamics, Fluid Mechanics, Numerical Analysis, and Thermodynamics

Awards: NASA Space Apps Challenge Global Nominee (Top 2%)

Experience

Stanford University School of Medicine

Visiting Researcher

CA, United States of America / [July 2026 – Present]

- Researching uncertainty quantification for 4D Flow MRI measurements under the supervision of Prof. Daniel B. Ennis
- Developing validation protocols and methods to assess the precision of MRI measurements across acquisition parameters

Agency for Defense Development (ADD)

Research Officers for National Defense Candidate

Daejeon, Republic of Korea / [Dec 2025 – Present]

- Selected for the elite national defense track (ROND), a joint-ministerial program hosted by the Ministry of National Defense and the Ministry of Science and ICT to cultivate the nation's top engineering talent
- Ranked as one of only 25 candidates selected nationwide, recognized by the Korean government for exceptional academic excellence and technical potential in Mechanical Engineering

Seoul National University

Undergraduate Researcher

Seoul, Republic of Korea / [Dec 2025 – Feb 2026]

- Developed a geometry-informed analysis framework to validate regions of interest and interpret MRI derived turbulence statistics under strong intravoxel shear and partial volume conditions
- Applied the workflow to gas turbine cooling internal flow configurations, using velocity field MRI features to infer intensive structures and asymmetry